3 TIME STEP CONTROL

3.1 Initial Time Value and User-Controlled Time Step, 200

This card is optional. See the description of each word on this card for the default values if this card is not entered.

W1(R) Initial time. If not entered, the simulation time at the start of the advancements is 0 for a NEW problem, the advancement time at the point of restart for a RESTART problem, or 0 for a RESTART problem in which the problem option switches from STDY-ST to TRANSNT or vice versa. If this card is entered, the simulation time is set to the entered value, which must be greater than or equal to 0. Setting the simulation time with this entry can be done on any NEW or RESTART problem but with most applications should only be used in NEW or RESTART problems that switch from the STDY-ST or TRANSNT options. See the cautions discussed in Section 2.9 for this capability. When needing to enter W2 but do not wish to enter a new initial time, enter -1.0, which is a flag to ignore this word.

W2(I) Control variable number for user-controlled time step. This word is optional. A nonzero number specifies a control variable whose value is used for user-specified time step control. The time step will be determined from the maximum of the value of the control variable and the current minimum time step entered on cards 201 through 299. The time step will be equal to or less than this value and depends on the current requested time step, the mass error and other error checks, the Courant limit, and the time-step reduction options.

3.2 Time Step Control, 201-299

At least one card of this series is required for NEW problems. If this series is entered for RESTART problems, it replaces the series from the problem being restarted. This series is not used for other problem types. Card numbers need not be consecutive.

W1(R) Time end for this set (s). This quantity must increase with increasing card number. The default action for the code is to stop at the first time step that is greater than or equal to this time. This means the code will probably not stop at the end time specified on this card, but will stop at some time larger than the end time. If the user wants to stop exactly at the end time, then the user needs to prefix the end time with a negative (-) sign.

W2(R) Minimum time step (s). The code starts out with this time step and increases by a factor 1.1 for the next step if the mass error is sufficiently low.

W3(R) Maximum time step (s). This quantity is also called the requested time step.

W4(I) Control option. This word has the packed format ssdtt. It is not necessary to input leading zeros.

ss. These digits represent a number from 0 through 15, are used to control the printed content of the major edits. The number is treated as a four-bit binary number. To decide how to set ss, pick some appropriate numbers from the set 0, 1, 2, 3, and 4, using the description below to decide which numbers to pick, then add the numbers together and set ss to that number.

ss = 0. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), all the standard major printed output is given.

ss = 1. OMIT HEAT STRUCTURE TEMPERATURE BLOCK OUTPUT. If the first bit from the right is set (i.e., ss = 1 if the other bits are not set), the heat structure temperature block is omitted.

ss = 2. OMIT SECOND PORTION OF THE JUNCTION BLOCK OUTPUT. If the second bit from the right is set (i.e., ss = 2 if the other bits are not set), the second portion of the junction block is omitted.

ss = 4. OMIT THIRD AND FOURTH PORTIONS OF VOLUME BLOCK OUTPUT. If the third bit from the right is set (i.e., ss = 4 if the other bits are not set), the third and fourth portions of the volume block is omitted.

ss = 8. OMIT STATISTICS BLOCK. If the fourth bit from the right is set (i.e., ss = 8 if the other bits are not set), the statistics block is omitted.

d. This digit represents a number from 0 through 7, can be used to obtain extra output at every hydrodynamic time step. The number is treated as a three-bit binary number. To decide how to set d, pick some appropriate numbers from the set 0, 1, 2, and 3, using the description below to decide which numbers to pick, then add the numbers together and set d to that number.

d = 0. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), the standard output at the requested frequency using the maximum time step is obtained (see words 5 and 6 of this card). If the number is nonzero, output is obtained at each successful time step; and the bits indicate which output is obtained. These options should be used carefully, since considerable output can be generated.

d = 1. MAJOR EDITS EVERY TIME STEP. If the first bit from the right is set (i.e., d = 1 if the other bits are not set), major edits are obtained every successful time step.

d = 2. MINOR EDITS EVERY TIME STEP. If the second bit from the right is set (i.e., d = 2 if the other bits are not set), minor edits are obtained every successful time step.

d = 4. PLOT RECORDS EVERY TIME STEP. If the third bit from the right is set (i.e., d = 4 if the other bits are not set), plot records are written every successful time step. As a practical example, assume you made a run with d = 0 and discovered that the plot points were too far apart to tell what was happening in the transient. One solution would be to decrease the plot frequency W5, to 1 to get a plot point for each requested time step and decrease the maximum time step size to get the desired detail in the plot. An easier solution would be to set ¨d = 4 and get plot points out each successful time step, even if they are smaller than the requested time step. For example, if tt = 3, then changingW4 from a 3 to a 403 would do the trick.

The digits tt, that represent a number from 0 through 63, are used to control the time step. The number is treated as a six-bit binary number. The effect of no bits being set, i.e., 0 being entered, and the effect of each bit are first described followed by the recommended combination of bits. To decide how to set tt, pick some appropriate numbers from the set 0, 1, 2, 4, 8, 16, and 32 using the description below to decide which numbers to pick, then add the numbers together and set tt to that number.

tt = 0. NORMAL OUTPUT. If no bits are set (i.e., the number is 0), no error estimate time step control is used, and the maximum time step is attempted for both hydrodynamic and heat structure advancement. The hydrodynamic time step, however, is reduced to the material Courant limit and further to the minimum time step for cases such as water property failures.

tt = 1. MASS ERROR CONTROLS THE TIME STEP. If the first bit from the right is set (i.e., tt = 1 if no other bits are set), the hydrodynamics advancement, in addition to the time step control when no bits are set, uses a mass error analysis to control the time step between the minimum and maximum time step.

tt = 2. HYDRODYNAMIC TIME STEP SAME AS HEAT CONDUCTION TIME STEP. If the second bit from the right is set (i.e., tt = 2 if the other bits are not set), the heat conduction/transfer time step is the same as the hydrodynamic time step; if the second bit from the right is not set, the heat conduction/transfer time step uses the maximum time step.

tt = 4. HEAT CONDUCTION AND HYDRODYNAMICS COUPLED IMPLICITLY. If the third bit from the right is set (i.e., tt = 4 if the other bits are not set), the heat conduction/transfer and hydrodynamics are coupled implicitly; if the third bit from the right is not set, the heat conduction/transfer and hydrodynamic advancements are done separately and the information between the models is coupled explicitly.

tt = 8. NEARLY-IMPLICIT TIME ADVANCEMENT. If the fourth bit from the right is set (i.e., tt = 8 if the other bits are not set), the nearly-implicit scheme is used to advance the hydrodynamics; if the fourth bit from the right is not set, the semi-implicit scheme is used to advance the hydrodynamics.

tt = 16. STEADY-STATE CONVERGENCE NOT CHECKED. If the fifth bit from the right is set (i.e., tt = 16 if the other bits are not set), the test for convergence of a steady-state calculation is not made; if the fifth bit from the right is not set, the test for convergence of a steady-state calculation is made.

tt = 32. AUTOMATIC SELECTION BETWEEN SEMI- AND NEARLY-IMPLICIT TIME ADVANCEMENT. If the sixth bit from the right is set (i.e., tt = 32 if the other bits are not set), the on-line algorithm selection of time migration is used to advance the hydrodynamics. The semi-implicit scheme will be used when the time step is below the Courant limit and the nearly-implicit will be used when a large time step can be taken.

We recommend not using tt equal to 0 except for special testing situations. The use of tt equal to 1 is possible if the maximum time step is kept sufficiently small to ensure that the explicit connection between the heat conduction/transfer and hydrodynamics calculations remains stable. If there is any doubt, use tt equal to or greater than 3 (sets first bit and second bit). Using tt equal to 3 or 11 specifies the semi-implicit or the nearly-implicit advancement scheme, respectively, with both schemes using time step control, the heat conduction and hydrodynamics use the same time step, and the heat conduction/transfer and hydrodynamics are advanced separately. Using tt equal to 7 or 15 specifies the same features as tt equal to 3 or 11 and, in addition, specifies the implicit advancement of the heat conduction/transfer with the hydrodynamics. We recommend the nearly-implicit scheme during a steady-state and/or self-initialization case problem where the time step is limited by the material Courant limit. The nearly-implicit scheme can also be used during slower phases of a transient problem, though we advise the user that the answers may change somewhat from the semi-implicit scheme answers (depending on the time step size). The nearly-implicit advancement scheme is still under development and assessment; most of the verification and assessment for the code has been done with the semi-implicit advancement scheme. We did not recommend use of the implicit coupling of the heat conduction/transfer and hydrodynamics in prior versions since the implicit coupling was only partially implemented. With the implicit coupling now complete, we encourage the use of tt equal to 7 or 15. Users should be cautioned that the implicit coupling is a recent addition to RELAP5 and is still under assessment. When using the implicit coupling, the heat conduction time step must be the same as the hydrodynamic time step. This requirement is currently not enforced by the coding. In steady-state calculations, setting the fifth bit (adding 16) for the early part of the run can ensure the calculation runs to a user-specified time; then, setting the fifth bit off can allow the steady-state convergence to test control the termination of the problem.

W5(I) Minor edit and plot frequency. This word has the packed format sssmmm. It is not necessary to input leading zeros. The mmm is the number of maximum or requested time advances per minor edit and the sss is the skip factor for the write of plot information. Thus sss*mmm is the number of maximum or requested time advances per plot-write on RSTPLT file. This feature enables to save the storage requirement for RSTPLT file. if no skip factor is set (sss=0), the default value is assigned, i.e. sss=1.

W6(I) Major edit frequency. This is the number of requested time-advances per major edit.

W7(I) Restart frequency. This is the number of requested time-advances per write of restart information.